Calibration of Normalized Label Probabilities for Prompt 4 (All Labels)

MMLU

	gpt-oss-20b	gpt-oss-120b	Ministral-8B-Instruct	Mistral-Nemo-Base	Mistral-Nemo-Instruct	Mistral-Small-3.1-24B-Base	Mistral-Small-3.2-24B-Instruct	Magistral-Small	Magistral-Small-Reasoning	Llama-3.1-70B	Llama-3.3-70B-Instruct	Llama-4-Scout-17B-16E	Llama-4-Scout-17B-16E-Instruct	Qwen3-30B-A3B-Base	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-R1-Distill-Llama-70B	DeepSeek-R1-Distill-Qwen-32B	gemma-3-27b-it
ECE	0.2167	0.2151	0.0483	0.0606	0.0881	0.0711	0.0456	0.0210	0.1900	0.0697	0.0620	0.0876	0.0682	0.0135	0.0770	0.1787	0.0844	0.0667	0.0735
AUROC	0.5000	0.5044	0.8387	0.7488	0.8439	0.8157	0.9226	0.9127	0.5592	0.8267	0.8963	0.8084	0.9276	0.9195	0.8781	0.5886	0.8352	0.8691	0.9161
Norm. Entropy of Bucket Counts	0.0001	0.0089	0.7885	0.8390	0.6475	0.7857	0.5539	0.6978	0.0753	0.6801	0.2875	0.3176	0.2683	0.6710	0.4119	0.0874	0.3209	0.3099	0.3616
Invalid Answer Count	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042
Accuracy	0.0000	0.0058	0.4814	0.4897	0.4844	0.6016	0.5939	0.5835	0.1050	0.5001	0.6149	0.4665	0.6171	0.5964	0.5378	0.1201	0.5143	0.5922	0.5847
Median of Label Probabilities	0.0000	0.0000	0.1132	0.1849	0.0398	0.1191	0.0178	0.0648	0.0000	0.0000	0.0000	0.0000	0.0000	0.0430	0.0000	0.0000	0.0000	0.0000	0.0001
IQR of Label Probabilities	0.0000	0.0000	0.3602	0.2948	0.3990	0.3771	0.4230	0.3818	0.0000	0.4225	0.4378	0.0000	0.5000	0.4223	0.4519	0.0000	0.0675	0.1480	0.4641
Median of Sum of Label Probabilities	0.0000	0.0000	0.5141	0.1221	0.8323	0.5655	0.2448	0.5483	0.0000	0.0156	0.0000	0.0017	0.0000	0.1112	0.0000	0.0000	0.0000	0.0002	0.0000
IQR of Sum of Label Probabilities	0.0000	0.0000	0.1643	0.0919	0.1437	0.1672	0.2384	0.2658	0.0000	0.0128	0.0000	0.0036	0.0000	0.1037	0.0000	0.0000	0.0001	0.0008	0.0000

ArcReasoning

	gpt-oss-20b	gpt-oss-120b	Ministral-8B-Instruct	Mistral-Nemo-Base	Mistral-Nemo-Instruct	Mistral-Small-3.1-24B-Base	Mistral-Small-3.2-24B-Instruct	Magistral-Small	Magistral-Small-Reasoning	Llama-3.1-70B	Llama-3.3-70B-Instruct	Llama-4-Scout-17B-16E	Llama-4-Scout-17B-16E-Instruct	Qwen3-30B-A3B-Base	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-R1-Distill-Llama-70B	DeepSeek-R1-Distill-Qwen-32B	gemma-3-27b-it
ECE	0.2167	0.2165	0.0488	0.0367	0.0196	0.0779	0.0305	0.0384	0.1587	0.0346	0.0198	0.0149	0.0175	0.0348	0.0237	0.2149	0.0678	0.0267	0.0199
AUROC	0.5000	0.5004	0.9911	0.9138	0.9787	0.9404	0.9979	0.9969	0.6252	0.9725	0.9870	0.9172	0.9901	0.9970	0.9919	0.5037	0.8867	0.9399	0.9958
Norm. Entropy of Bucket Counts	0.0000	0.0008	0.5996	0.7498	0.4647	0.6351	0.3107	0.4455	0.1268	0.4986	0.2292	0.2035	0.2176	0.3618	0.2612	0.0052	0.3619	0.3572	0.2398
Invalid Answer Count	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358
Accuracy	0.0000	0.0009	0.7052	0.6828	0.6775	0.7421	0.7439	0.7382	0.2222	0.6894	0.7406	0.6263	0.7362	0.7466	0.7370	0.0048	0.5277	0.6307	0.7362
Median of Label Probabilities	0.0000	0.0000	0.0398	0.1153	0.0058	0.0454	0.0033	0.0124	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
IQR of Label Probabilities	0.0000	0.0000	0.3750	0.2887	0.4371	0.3428	0.4511	0.4183	0.0000	0.4378	0.4999	0.0000	0.5000	0.4603	0.4378	0.0000	0.1848	0.3208	0.4644
Median of Sum of Label Probabilities	0.0000	0.0000	0.5648	0.1214	0.8138	0.5092	0.4134	0.6448	0.0000	0.0127	0.0000	0.0012	0.0000	0.1425	0.0000	0.0000	0.0000	0.0002	0.0000
IQR of Sum of Label Probabilities	0.0000	0.0000	0.1223	0.1340	0.1550	0.1944	0.2383	0.2130	0.0001	0.0072	0.0000	0.0015	0.0000	0.0847	0.0000	0.0000	0.0001	0.0009	0.0000

GSM8KMC

	gpt-oss-20b	gpt-oss-120b	Ministral-8B-Instruct	Mistral-Nemo-Base	Mistral-Nemo-Instruct	Mistral-Small-3.1-24B-Base	Mistral-Small-3.2-24B-Instruct	Magistral-Small	Magistral-Small-Reasoning	Llama-3.1-70B	Llama-3.3-70B-Instruct	Llama-4-Scout-17B-16E	Llama-4-Scout-17B-16E-Instruct	Qwen3-30B-A3B-Base	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-R1-Distill-Llama-70B	DeepSeek-R1-Distill-Qwen-32B	gemma-3-27b-it
ECE	0.2166	0.2095	0.0851	0.1058	0.0981	0.1050	0.0330	0.0585	0.1918	0.2030	0.1505	0.1739	0.1678	0.0210	0.2297	0.2112	0.1855	0.0651	0.1208
AUROC	0.5002	0.5147	0.6546	0.5937	0.6907	0.7007	0.8293	0.7592	0.5507	0.5487	0.8199	0.6690	0.8206	0.9185	0.5985	0.5116	0.6007	0.8377	0.8479
Norm. Entropy of Bucket Counts	0.0005	0.0167	0.8426	0.7471	0.8349	0.8365	0.8351	0.8467	0.0689	0.1087	0.4120	0.2960	0.3236	0.7857	0.4601	0.0146	0.1342	0.2421	0.4939
Invalid Answer Count	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468
Accuracy	0.0000	0.0020	0.2866	0.3535	0.2895	0.4796	0.3907	0.3851	0.1031	0.0770	0.4464	0.2425	0.4543	0.5216	0.1638	0.0165	0.1727	0.5740	0.4120
Median of Label Probabilities	0.0000	0.0000	0.1892	0.2632	0.1661	0.2105	0.1541	0.1767	0.0000	0.0000	0.0000	0.0000	0.0000	0.1144	0.0000	0.0000	0.0000	0.0000	0.0035
IQR of Label Probabilities	0.0000	0.0000	0.2837	0.2871	0.3584	0.3598	0.3491	0.3035	0.0000	0.0000	0.4688	0.0000	0.5000	0.4302	0.3775	0.0000	0.0000	0.0000	0.4668
Median of Sum of Label Probabilities	0.0000	0.0000	0.2930	0.0790	0.9314	0.6450	0.1432	0.2768	0.0000	0.0000	0.0006	0.0005	0.0000	0.1045	0.0000	0.0000	0.0000	0.0001	0.0000
IQR of Sum of Label Probabilities	0.0000	0.0000	0.0855	0.0356	0.0435	0.1498	0.1258	0.1642	0.0000	0.0000	0.0140	0.0011	0.0001	0.0863	0.0000	0.0000	0.0000	0.0002	0.0001

GPQA

	gpt-oss-20b	gpt-oss-120b	Ministral-8B-Instruct	Mistral-Nemo-Base	Mistral-Nemo-Instruct	Mistral-Small-3.1-24B-Base	Mistral-Small-3.2-24B-Instruct	Magistral-Small	Magistral-Small-Reasoning	Llama-3.1-70B	Llama-3.3-70B-Instruct	Llama-4-Scout-17B-16E	Llama-4-Scout-17B-16E-Instruct	Qwen3-30B-A3B-Base	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-R1-Distill-Llama-70B	DeepSeek-R1-Distill-Qwen-32B	gemma-3-27b-it
ECE	0.2183	0.2144	0.0933	0.1283	0.1546	0.1221	0.1472	0.1185	0.2167	0.2119	0.2133	0.2104	0.2493	0.0859	0.2372	0.2066	0.2133	0.1768	0.2429
AUROC	0.5004	0.5082	0.5709	0.5409	0.5912	0.5799	0.6515	0.6260	0.5007	0.5408	0.6902	0.5617	0.6794	0.6566	0.5750	0.5238	0.5533	0.6255	0.6319
Norm. Entropy of Bucket Counts	0.0094	0.0169	0.7979	0.7787	0.8416	0.8068	0.8165	0.8465	0.0032	0.2930	0.3935	0.2952	0.3124	0.8389	0.5837	0.0297	0.1642	0.1726	0.5571
Invalid Answer Count	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448
Accuracy	0.0000	0.0022	0.2344	0.3080	0.2879	0.3259	0.3304	0.3214	0.0022	0.0915	0.3571	0.1004	0.3393	0.2589	0.1540	0.0379	0.0781	0.2232	0.2277
Median of Label Probabilities	0.0000	0.0000	0.2182	0.2582	0.1657	0.2518	0.1354	0.1794	0.0000	0.0000	0.0000	0.0000	0.0000	0.2164	0.0000	0.0000	0.0000	0.0000	0.0140
IQR of Label Probabilities	0.0000	0.0000	0.1961	0.3362	0.3317	0.3395	0.3405	0.2869	0.0000	0.0000	0.4189	0.0000	0.5000	0.2583	0.4688	0.0000	0.0000	0.0000	0.4601
Median of Sum of Label Probabilities	0.0000	0.0000	0.3485	0.1913	0.7377	0.4572	0.1237	0.2796	0.0000	0.0000	0.0000	0.0000	0.0001	0.1067	0.0000	0.0000	0.0000	0.0000	0.0000
IQR of Sum of Label Probabilities	0.0000	0.0000	0.1319	0.1299	0.1681	0.1572	0.1771	0.2362	0.0000	0.0049	0.0000	0.0004	0.0025	0.1145	0.0000	0.0000	0.0000	0.0000	0.0001